

Scratch Model Summary

Model Idea:

Scratches are created on the surface of pebbles as after the front of the running band has passed over them, the scratch is created due to the rotation of the curling stone as it moves. Therefore, the scratches are created at an angle tangent to the running band at the point it touches the pebble.

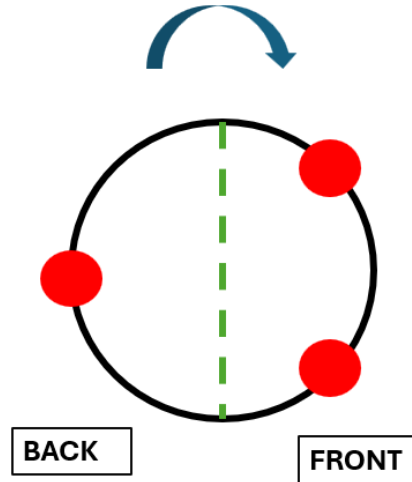


Figure 1: A diagram illustrating the curling stone, rotating in a clockwise direction, split into front and back sections with 2 pebbles touching the front of the running band and 1 touching the back.

This scratch influences the back of the running band creating a force perpendicular to the curling stone's running band. The equation for this force is shown by the equation:

$$F = \kappa \times \text{Friction}$$

Where $\kappa = A \cos(\phi)$ and is always a positive value, ϕ is the angle of the scratch, and A is a constant with value $0.07 \leq A \leq 0.21$. The Friction value is μmg .

The constant A is directly proportional to the diameter of the pebble and indirectly proportional to the spacing between each scratch. A larger diameter leads to a greater force, and smaller spacing also leads to a greater force.

The equations used in this model are adapted from (Penner, 2019).

Assumptions:

Ignoring the height of pebbles.

Constant coefficient of friction for the rotation of the stone.

Asperities on the running band of the stone do not need to be individually accounted for.

Ignoring scratches on the pebbles at the back of the running stone.

Predictions:

Predict the curling stone to curl in the same direction as its direction of rotation.

Predict the curling stone to have a curl greater than a few cm and less than 2m, and for its curl to be directly proportional to the constant A.

Results:

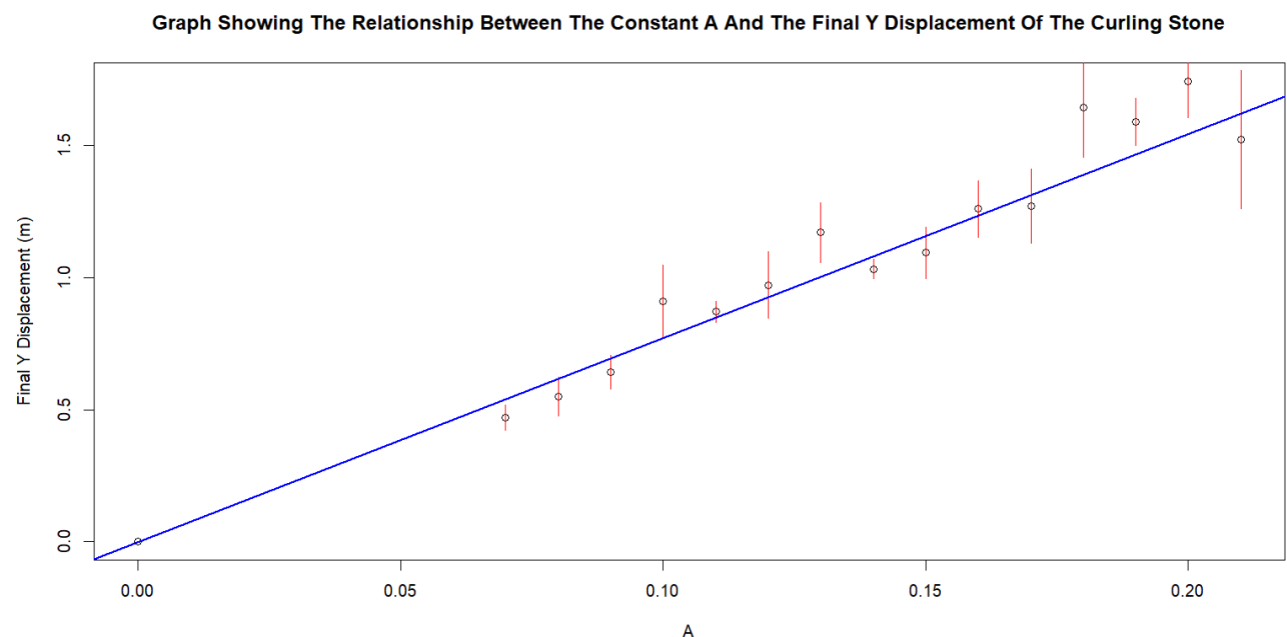


Figure 2: A graph showing Y Displacement against the constant A, results were recorded 6 times for each value of A and an average was taken and plotted on the graph with the standard deviation as the error bars.

This graph shows that the curl of the curling stone can be predicted but as the constant A is increased, there is more uncertainty in the prediction. It also shows that in this model, the final y displacement of the curling stone is directly proportional to the constant A.

Conclusion:

Overall, the scratch model produces a curl between 0.5 and 2m which is as predicted. The magnitude of the scratch is proportional to the diameter of the pebble and indirectly proportional to the scratch spacing. This model also correctly predicts the magnitude of the curl; a clockwise rotation of the curling stone produces a negative Y displacement.

Future Work:

- Accounting for thermal effects
- Accounting for pivoting motion